COMPARATIVE EVALUATION OF DECISION TREE AND RANDOM FOREST ALGORITHMS FOR STROKE RISK PREDICTION USING A SENSITIVITY-BASED APPROACH

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Abstract

Stroke prediction from routine clinical data is hindered by severe class imbalance, with confirmed events representing fewer than five percent of most population-level datasets. Standard accuracy-optimized pipelines systematically underdetect minority-class events, limiting clinical utility. This study presents a sensitivity-centered comparative evaluation of Decision Tree (CART) and Random Forest algorithms on a public stroke dataset of 5,110 observations (1:19 class ratio). Three progressive scenarios — baseline, SMOTE-augmented, and recall-optimized with nested cross-validation — were assessed under a strict three-way data split (60/20/20%) that completely isolated the test set from all modeling decisions, including threshold selection. Probability thresholds were derived exclusively from the validation set using a minimum-recall constraint of 0.70, eliminating look-ahead bias. Post-hoc evaluation included bootstrap confidence intervals (1,000 iterations), McNemar testing, calibration analysis, risk stratification, and clinical workload projection. The recall-optimized CART model achieved 0.86 sensitivity (7 false negatives from 50 stroke cases), a Brier Score of 0.1501, and a 7.85-fold relative risk gradient across probability strata. The optimized Random Forest attained marginally higher recall (0.92) but generated 532 false positives versus 406 for CART, reducing accuracy to 47.55% versus 59.59% and imposing a substantially heavier referral burden. McNemar's test confirmed this difference in classification balance is statistically significant ($p < 0.001$). Within this experimental context, CART Optimized indicates a more balanced trade-off between sensitivity and false positive workload under capacity-constrained screening conditions. These findings suggest that leakage-safe threshold optimization combined with nested cross-validation produces a clinically viable screening framework superior to standard accuracy-maximizing approaches.

Keywords : *Decision Tree, Machine Learning, Random Forest, Sensitivity, Stroke.*

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1. INTRODUCTION

Cerebrovascular accidents, commonly known as strokes, represent a major global public health challenge. The World Stroke Organization estimates that one in four adults over the age of 25 will experience a stroke in their lifetime, and the condition accounts for approximately 11% of all deaths worldwide [1]. Beyond mortality, stroke survivors frequently face lasting neurological impairment, cognitive decline, and long-term care dependency, all of which impose enormous economic burdens on both households and healthcare systems [1], [2]. Identifying individuals at elevated risk before an acute event occurs is therefore central to effective prevention: early risk stratification enables targeted intervention, lifestyle counseling, and prophylactic pharmacotherapy that can substantially reduce event probability and severity [3].

The rapid accumulation of digitized electronic health records has opened significant opportunities for data-driven risk stratification. Machine learning methods are particularly appealing in this context because they can detect complex, nonlinear interactions among clinical variables that conventional statistical models often miss [3], [4]. However, a persistent structural obstacle limits the

direct application of standard classification algorithms to stroke prediction: class imbalance. In population-level screening datasets, confirmed stroke cases typically make up fewer than five percent of all records, yielding class ratios of 1:19 or more extreme [5], [6]. Under such conditions, classifiers trained to minimize global loss functions tend to predict the majority class almost exclusively, achieving high nominal accuracy while failing at the clinically essential task of detecting rare but severe minority-class events [7].

This concern is amplified by the asymmetric cost structure inherent to stroke screening. A false negative, defined here as an undetected high-risk individual, may go without intervention until an acute event occurs, potentially resulting in permanent disability or death. A false positive, by contrast, triggers a follow-up clinical assessment: a real inconvenience, but one that is correctable within a two-stage screening workflow [8]. This asymmetry means that stroke prediction models must be evaluated and optimized primarily on sensitivity and false negative rate, rather than on aggregate accuracy, ROC-AUC, or F1-score averaged across both classes.

Prior research has explored a range of technical strategies for improving minority-class detection in imbalanced medical datasets, including oversampling via SMOTE, cost-sensitive learning, ensemble methods, and threshold adjustment [9],[10],[11]. A recurring methodological gap in the stroke prediction literature, however, is the continued reliance on aggregate performance indicators, such as overall accuracy, macro-averaged F1, or area under the ROC curve, as primary selection criteria, while sensitivity is treated as a secondary metric reported for completeness rather than as a driver of model choice [12]. This practice risks endorsing models that appear statistically sound under conventional benchmarks but are fundamentally unsuited for clinical deployment.

A further, less-discussed methodological hazard is the common conflation of validation and test data in threshold optimization [13]. Many published studies select classification thresholds by inspecting test-set performance curves, a practice that constitutes look-ahead data leakage and systematically overstates the sensitivity achievable in prospective application [14]. Likewise, hyperparameter tuning conducted on the training partition without nested cross-validation can introduce optimistic bias in reported generalization performance [15].

The present study addresses these shortcomings through a structured evaluation of two widely used tree-based classifiers, CART and Random Forest, for stroke risk prediction. The methodological contributions are threefold: (1) a three-way data partition that enforces complete separation of validation and test sets, with thresholds selected exclusively from validation data; (2) a nested cross-validation architecture that cleanly separates hyperparameter tuning from generalization estimation; and (3) a comprehensive post-hoc evaluation suite covering bootstrap confidence intervals, McNemar statistical testing, calibration curves, risk stratification, false negative profiling, and clinical workload projection [16]. The novelty of this framework lies not in the algorithms themselves but in the systematic prioritization of sensitivity as the primary criterion at every stage of model development and evaluation. This approach directly aligns machine learning optimization with the demands of clinical screening.

2. METHOD

This study uses a quantitative experimental design grounded in reproducible machine learning methodology. Figure 1 provides an overview of the complete research pipeline, which proceeds through five sequential stages: data acquisition and preprocessing, stratified three-way partitioning, scenario-based model training under nested cross-validation, validation-set threshold selection, and final test-set evaluation with comprehensive statistical reporting [17].

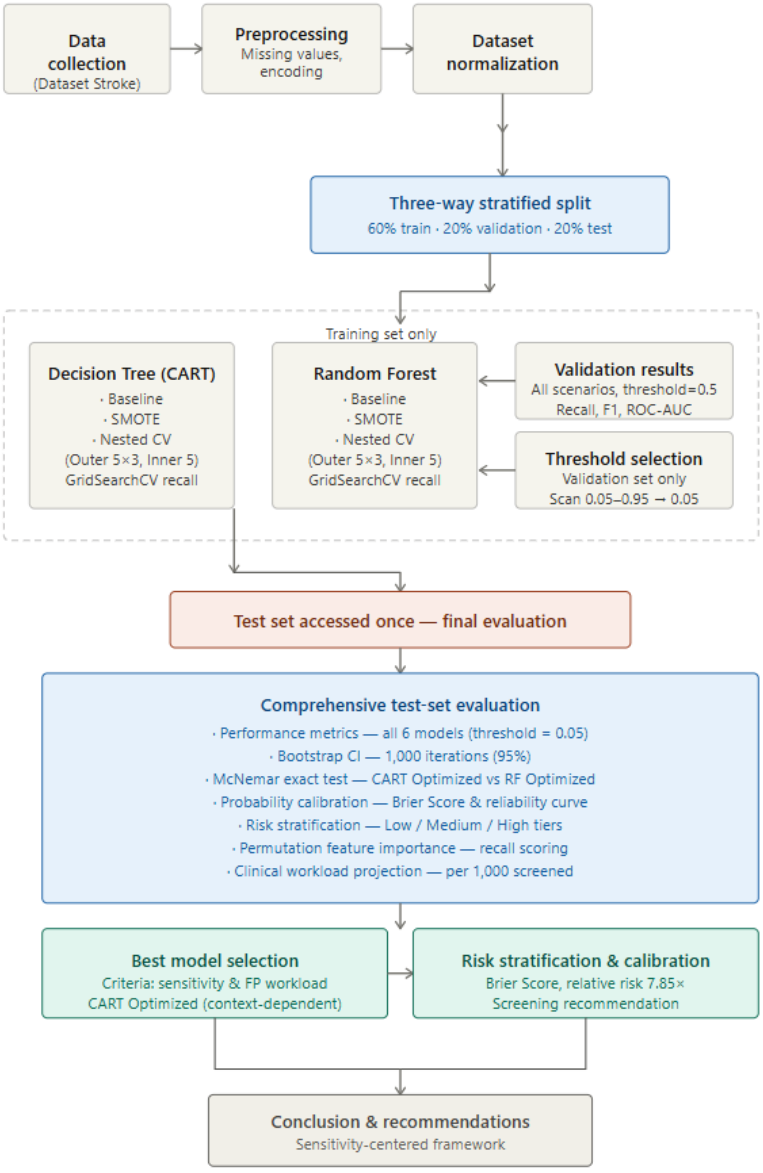


Figure 1. Block Diagram Representing the Flow of Research

2.1 Dataset and Data Characteristics

The dataset is a publicly available stroke health record deposited on Kaggle, originally compiled from anonymized patient records [18]. It contains 5,110 observations, each described by ten predictor variables and one binary outcome indicating stroke occurrence. Predictors span four domains: demographic attributes (age, sex), clinical conditions (hypertension, heart disease), social determinants (marital status, work type, residential setting), and lifestyle factors (smoking status, average glucose level, body mass index). Table 1 presents a representative excerpt of the raw dataset.

Tabel 1. Dataset used in the Research

No	Gender	Age	Hypertension	Heart Disease	Ever Married	Work Type	Residence	Avg Glucose	BMI	Smoking Status	Stroke
1	Male	67	0	1	Yes	Private	Urban	228.69	36.06.00	Formerly Smoked	1
2	Female	61	0	0	Yes	Self-employed	Rural	202.21.00	NaN	Never Smoked	1
3	Male	80	0	1	Yes	Private	Rural	105.92	32.05.00	Never Smoked	1
4	Female	49	0	0	Yes	Private	Urban	171,23	34,4	smokes	1
5	Female	79	1	0	Yes	Self-employed	Rural	174,12	24	never smoked	1
6	Male	81	0	0	Yes	Private	Urban	186,21	29	formerly smoked	1
7	Male	74	1	1	Yes	Private	Rural	70,09	27,4	never smoked	1
8	Female	69	0	0	No	Private	Urban	94,39	22,8	never smoked	1
9	Female	59	0	0	Yes	Private	Rural	76,15		Unknown	1
10	Female	78	0	0	Yes	Private	Urban	58,57	24,2	Unknown	1
...	...	...	...	...	...	...	...	...	...	...	...
5110	Female	44	0	0	Yes	Govt_job	Urban	85.28.00	26.02.00	Unknown	0

The The class distribution is heavily skewed: 249 observations (4.87%) correspond to confirmed stroke events, while the remaining 4,861 (95.13%) are non-stroke cases, yielding an approximate 1:19 minority-to-majority ratio [19]. This degree of imbalance poses a severe challenge for standard classification algorithms that optimize cross-entropy or Gini impurity, as these objectives assign disproportionate weight to correctly classifying the dominant class.

2.2 Data Preprocessing

Three preprocessing steps were applied prior to model training. First, missing values in the body mass index variable, totaling 201 observations (3.93% of the dataset), were imputed using the column median. Median imputation was preferred over mean imputation because the right-skewed BMI distribution (skewness $\approx$ 1.055) makes the mean susceptible to distortion by high-leverage outliers, while the median provides a more robust central tendency estimate [20]. Deletion was explicitly rejected, as removing 201 observations would further shrink the already limited 249-case stroke minority.

Second, the three continuous predictors (age, average glucose level, and BMI) were standardized using z-score normalization (StandardScaler), centering each variable at zero mean and unit variance. This prevents features with large numeric scales from dominating computations during tree construction and cross-validation scoring [21]. Third, five categorical variables (gender, marital status, work type, residence type, and smoking status) were converted to binary indicator columns via One-Hot Encoding, expanding the feature space from 10 to 21 dimensions and avoiding the imposition of artificial ordinal structure [22]. Table 2 summarizes each preprocessing decision with its statistical justification.

Tabel 2. Preprocessing Decisions and Justifications

Stage	Variable(s) Affected	Method	Statistical Justification
Missing Value Imputation	BMI (201 values, 3.93%)	Median Imputation	Skewness ≈1.055; median robust to extreme values; deletion would reduce minority class further
Feature Normalization	Age, average glucose level, BMI	StandardScaler (μ=0, σ=1)	Ensures equal scale contribution across features; required for stable cross-validation scoring
Categorical Encoding	Gender, marital status,	One-Hot	Avoids ordinal assumptions;

	work type, residence, smoking status	Encoding	expands feature space from 10 to 21 dimensions

Tabel 3. Sample Dataset After Preprocessing

No	Gender	Age	Hypertension	Heart Disease	Ever Married	Work Type	Residence	Avg Glucose	BMI	Smoking Status	Stroke
1	Male	67	0	1	Yes	Private	Urban	228.69	36.06.00	Formerly Smoked	1
2	Female	61	0	0	Yes	Self-employed	Rural	202.21.00	28.01.00	Never Smoked	1
3	Male	80	0	1	Yes	Private	Rural	105.92	32.05.00	Never Smoked	1
4	Female	49	0	0	Yes	Private	Urban	171,23	34,4	smokes	1
5	Female	79	1	0	Yes	Self-employed	Rural	174,12	24	never smoked	1
6	Male	81	0	0	Yes	Private	Urban	186,21	29	formerly smoked	1
7	Male	74	1	1	Yes	Private	Rural	70,09	27,4	never smoked	1
8	Female	69	0	0	No	Private	Urban	94,39	22,8	never smoked	1
9	Female	59	0	0	Yes	Private	Rural	76,15	28,1	Unknown	1
10	Female	78	0	0	Yes	Private	Urban	58,57	24,2	Unknown	1
...	...	...	...	...	...	...	...	...	...	...	...
5110	Female	44	0	0	Yes	Govt_job	Urban	85.28.00	26.02.00	Unknown	0

2.3 Three-Way Stratified Data Partition

A core methodological constraint of this study was the strict separation of three non-overlapping data subsets: a training set (60%), a validation set (20%), and a held-out test set (20%). All splits were stratified on the stroke label to preserve the original 1:19 class ratio across each partition. The training set was used exclusively for model fitting and SMOTE augmentation. The validation set served as the sole basis for threshold optimization and scenario comparison, while the test set was reserved exclusively for final model evaluation [23].

This three-way partitioning strategy was adopted to ensure strict isolation between model development, threshold selection, and final evaluation stages. Performing threshold optimization directly on the test set introduces look-ahead bias, which can artificially inflate reported sensitivity and compromise the reliability of performance estimates. By introducing a dedicated validation partition between the training and testing phases, the framework ensures that threshold selection remained fully independent of the final evaluation data, thereby reducing the risk of overfitting during performance optimization. The resulting split yielded 3,066 training observations (149 stroke cases), 1,022 validation observations (50 stroke cases), and 1,022 test observations (50 stroke cases).

2.4 Experimental Scenarios

Model performance was examined across three progressively refined experimental conditions, summarized in Table 4. The baseline scenario trained both algorithms with default scikit-learn hyperparameters and no imbalance correction, establishing a reference that isolates the effect of class skew on classification behavior. The SMOTE scenario applied Synthetic Minority Over-sampling Technique exclusively within the training pipeline — preventing synthetic samples from contaminating the validation or test partitions — while retaining default hyperparameters to isolate the resampling contribution from that of tuning. The optimization scenario combined SMOTE with nested cross-validation and recall-oriented GridSearchCV to jointly address imbalance, model selection bias, and sensitivity maximization.

Tabel 4. Description of Experimental Modeling Scenarios

Scenario	Algorithm	Imbalance Handling	Hyperparameter Strategy	Primary Objective
Baseline	CART, RF	None	Default parameters	Reference: isolate imbalance effect
SMOTE	CART, RF	SMOTE (training set only)	Default parameters	Isolate resampling contribution
Optimized	CART, RF	SMOTE (training set only)	Nested CV + GridSearchCV (recall scoring)	Maximize sensitivity; minimize false negatives

2.5 Classification Algorithms

Decision Tree (CART) was selected as the primary interpretable classifier. CART recursively partitions the feature space by selecting, at each node, the binary split that maximizes the reduction in Gini impurity [24]. For a node containing n samples with class proportions p_0 and $p_1 = 1 - p_0$, the Gini criterion is:

$$Gini = 1 - \sum_i p_i^2 = 2p_0p_1 \qquad (1)$$

Random Forest was included as the ensemble baseline. It aggregates predictions from B independently grown trees through majority voting, with each tree trained on a bootstrap sample of the training data. At each split, only a random subset of √m features is considered, introducing two complementary sources of variance reduction: bagging and feature randomization [25]. The ensemble prediction probability for class k is:

$$\hat{P}(Y = k|x) = \frac{1}{B}\sum_{t=1}^{B} \hat{P}(Y = k|x) \qquad (2)$$

The single-tree structure of CART makes it tractable for clinical audit — a physician can trace any prediction to an explicit decision path — whereas Random Forest trades transparency for variance reduction.

2.6 Nested Cross-Validation Architecture

To reduce the risk of optimistic bias during hyperparameter optimization, this study implemented a fully nested cross-validation framework. The outer evaluation loop employed RepeatedStratifiedKFold with five folds and three repetitions (15 total iterations), providing a more stable estimate of model generalization performance. Within each outer training partition, an inner five-fold StratifiedKFold procedure was passed to GridSearchCV for hyperparameter selection by maximizing mean cross-validation recall. This separation between the inner optimization stage and the outer evaluation stage ensures that information from the outer test fold does not influence

hyperparameter selection, thereby preventing the optimistic bias commonly associated with single-loop validation procedures [26].

The parameter search space for CART included maximum tree depth ∈ {None, 5, 10, 20}, minimum samples per split ∈ {2, 5, 10}, and minimum samples per leaf ∈ {1, 2, 5}. For Random Forest, the evaluated parameters included the number of estimators ∈ {100, 200}, maximum depth ∈ {None, 10, 20}, and minimum samples per split ∈ {2, 5}. After nested cross-validation established performance estimates on the training data, each GridSearchCV configuration was refitted using the full training partition to produce the final model used for validation-set threshold optimization and subsequent test-set evaluation [27].

2.7 Evaluation Metrics and Threshold Selection

Classification performance was quantified using a comprehensive metric suite. The primary criteria were Recall (Sensitivity) and false negative count; secondary criteria included Precision, F1-score, Accuracy, ROC-AUC, PR-AUC, and Brier Score. Formal definitions are given in Equations (1)-(6):

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

$$Precision = \frac{TP}{TP + FP} \tag{4}$$

$$F1 = \frac{2\ (Precision \times Recall)}{Precision + Recall} \tag{5}$$

$$Accuracy = \frac{TN}{TN + FP} \tag{6}$$

$$ROC - AUC = \int_0^1 TPR(\ FPR) d(FPR) \tag{7}$$

$$Brier\ Score = \frac{1}{n} \sum_{i=1}^{n} (p_i - y_i)^2 \tag{8}$$

Threshold optimization was conducted exclusively on the validation set, scanning candidate thresholds from 0.05 to 0.95 in steps of 0.05. The selection criterion enforced a minimum recall of 0.70, reflecting a clinical constraint that no more than 30% of true stroke cases may be missed in a first-stage screening application [28]. Among all thresholds meeting this constraint, the one yielding the fewest false negatives was selected. The chosen threshold was then applied unchanged to all subsequent test-set evaluations, preserving complete isolation of the test partition [29].

2.8 Confidence Intervals and Statistical Testing

Given the limited number of stroke cases in the test partition (n = 50), point estimates alone are insufficient for robust inference. Bootstrap resampling with 1,000 iterations was therefore applied to produce 95% confidence intervals for all primary metrics on the test set. In each iteration, a sample of equal size was drawn with replacement from the test observations, and metrics were recomputed using the chosen threshold; the 2.5th and 97.5th percentiles of the resulting distribution define the interval bounds. McNemar's exact test was applied to the 2×2 contingency table of correct versus incorrect predictions for the two optimized models, providing a formal test of the null hypothesis that the models commit classification errors at equal rates [30].

3. RESULT

3.1 Baseline Model Performance

Baseline performance was evaluated on the held-out test set using probability thresholds derived exclusively from the validation partition. A comprehensive comparison of all experimental scenarios is provided in Table 5 (Section 3.5). Under default hyperparameters and without imbalance correction, the baseline CART model achieved an overall accuracy of 90.70% but identified only 6 of 50 stroke cases, resulting in a recall of 0.12 and 44 false negatives. In contrast, the baseline Random Forest model attained a recall of 0.74 by correctly detecting 37 stroke cases, although its accuracy decreased to 71.82% due to the substantially larger number of false positives generated at the selected operating threshold. These findings illustrate the commonly observed accuracy–sensitivity trade-off in severely imbalanced classification tasks. Without explicit recall-oriented optimization, classifiers either favor the majority class and achieve high overall accuracy (CART) or improve minority-class detection at the expense of increased false positive burden (Random Forest). In both cases, the baseline configurations demonstrate limited suitability for sensitivity-oriented stroke screening applications.

Confusion matrices for all experimental scenarios are presented in Figure 2. The baseline results further indicate that both algorithms remain strongly influenced by the underlying class imbalance prior to the application of resampling and optimization procedures.

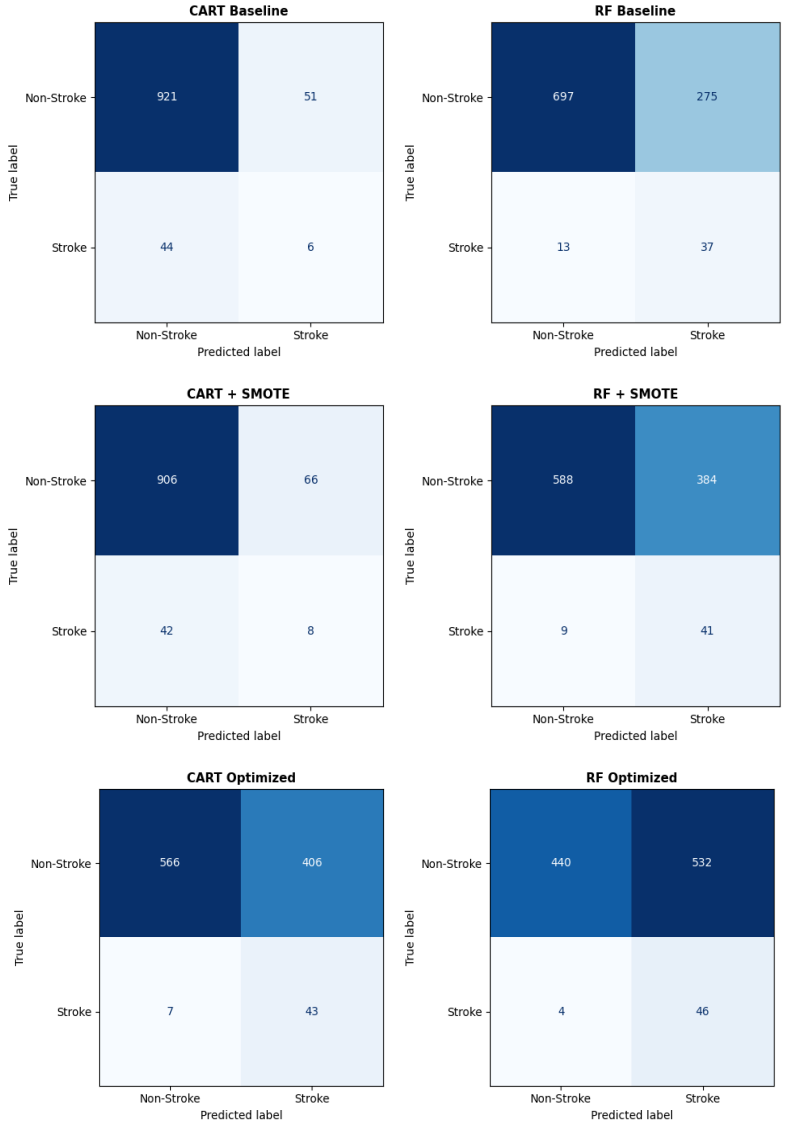


Figure 2. Confusion Matrices Across All Experimental Configurations Evaluated on the Test Set

3.2 Effect of SMOTE Augmentation

Applying SMOTE within the training pipeline produced divergent effects across the two classifiers, and neither model achieved sensitivity levels comparable to the optimized scenario. For CART, SMOTE resulted in only a modest increase in recall from 0.12 to 0.16, while overall classification performance deteriorated. False positives increased from 51 to 66, accuracy declined from 90.70% to 89.43%, and 42 stroke cases remained undetected. This pattern is consistent with SMOTE-induced overfitting, where improved sensitivity on resampled training data does not generalize effectively to naturally distributed validation and test observations. For Random Forest, SMOTE produced a more substantial improvement in minority-class detection. Recall increased from 0.74 to 0.82, while false negatives decreased from 13 to 9. However, this gain was accompanied by a considerable rise in false positives (384 versus 275), reducing overall accuracy to 61.55%. Collectively, these findings suggest that resampling alone, without explicit threshold optimization, does not consistently produce a stable operating point under severe class imbalance conditions.

The differing responses of CART and Random Forest can be explained by their underlying learning characteristics. SMOTE generates synthetic minority-class samples through linear interpolation between existing minority observations in feature space. Under the original 1:19 class ratio, many synthetic samples are likely to occupy boundary regions where minority and majority distributions overlap. As a high-variance single-tree classifier, CART is particularly sensitive to these boundary perturbations and may construct highly specific decision rules that generalize poorly beyond the resampled training distribution. Random Forest is comparatively more robust to this effect. Although individual trees may overfit to synthetic observations, the ensemble aggregation process reduces variance amplification through bagging and majority voting, thereby smoothing localized boundary artifacts introduced during resampling.

3.3 Nested Cross-Validation Results

The nested cross-validation performance estimates for both optimized models across the 15 outer-loop iterations (five folds × three repetitions) are summarized below. Table 5 presents the final held-out test-set results across all experimental scenarios, while Table 6 reports the corresponding bootstrap confidence intervals; both are discussed further in Sections 3.5 and 3.6. The optimized CART model achieved a mean outer-loop recall of 0.616 ($\sigma = 0.113$), with a 95% confidence interval ranging from 0.402 to 0.788. The relatively wide interval reflects the inherent variability of recall estimation when each outer fold contains only approximately 25 stroke cases. Random Forest exhibited a substantially lower mean recall of 0.235 ($\sigma = 0.077$, 95% CI [0.112–0.387]), consistent with its reduced minority-class sensitivity across all experimental scenarios. The consistent gap between CART and RF across all 15 iterations confirms that the recall advantage is a stable property of the recall-oriented optimization and not an artifact of a single favorable data partition. Figure 3 visualizes the fold-by-fold recall trajectory for both models.

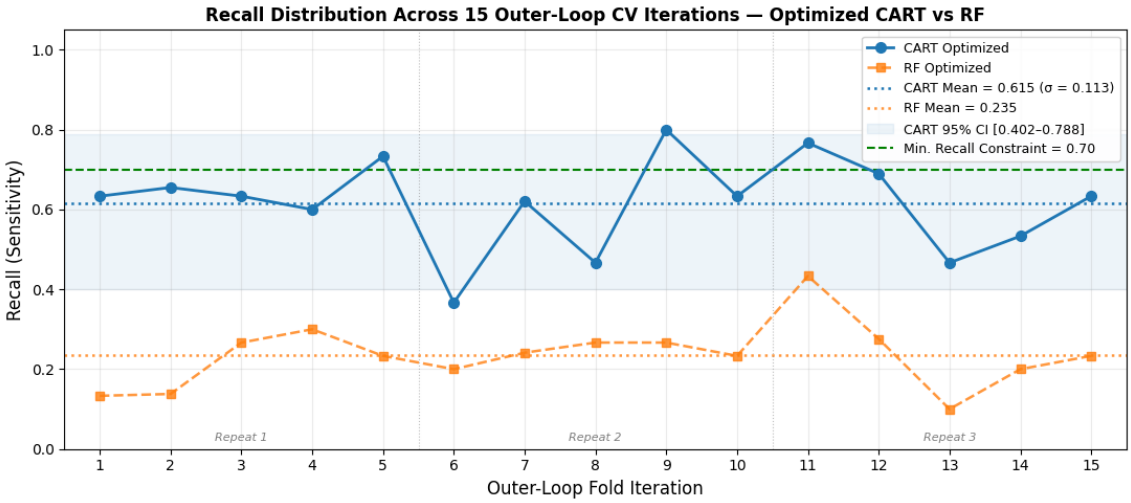


Figure 3. Recall Performance Across 15 Outer-Loop Cross-Validation Iterations for Optimized CART and Random Forest

3.4 Validation-Set Threshold Selection

Threshold optimization was conducted exclusively on the validation set by scanning candidate thresholds from 0.05 to 0.95 in increments of 0.05. The predefined minimum recall target of 0.70 was first satisfied at a threshold of 0.05. Among all thresholds meeting this criterion, 0.05 produced the smallest false negative count and was therefore selected as the operating threshold for all subsequent test-set evaluations. At this threshold, the validation-set recall reached 0.88, with 6 false negatives and 418 false positives, yielding a precision of 0.095. All threshold-selection procedures and associated performance estimates reported in this subsection were derived exclusively from the validation partition. No threshold decision was informed by observations from the held-out test set. Figure 4 illustrates the false-negative versus false-positive trade-off curve on the validation set, with the selected operating point annotated.

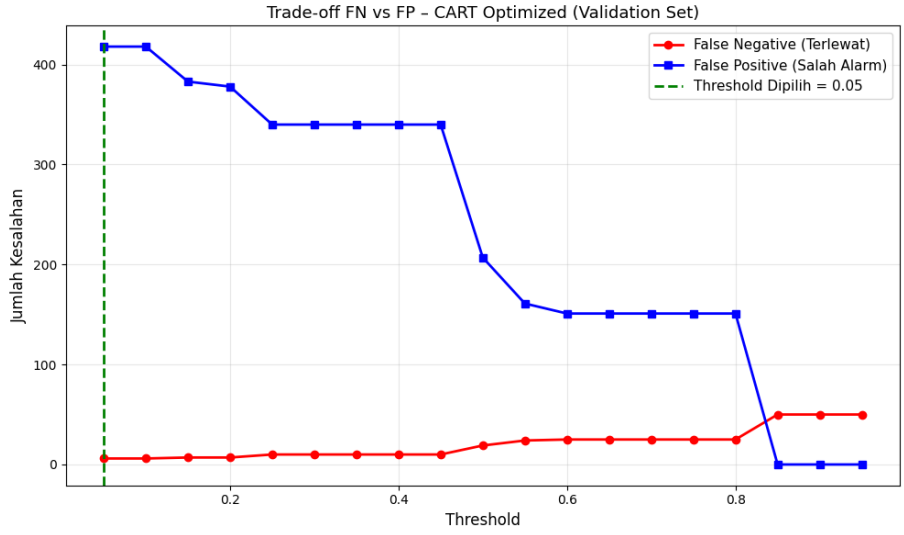


Figure 4. Threshold Trade-Off Curve

3.5 Comprehensive Test-Set Results

Tabel 5. Complete Performance Comparison Across All Experimental Scenarios

Model	Scenario	Acc. (%)	Precision	Recall	F1	ROC-AUC	Brier Score	False Neg.

CART	Baseline	90.70	0.105	0.12	0.112	0.534	—	44
RF	Baseline	71.82	0.119	0.74	0.204	0.778	—	13
CART	SMOTE	89.43	0.108	0.16	0.129	0.546	—	42
RF	SMOTE	61.55	0.096	0.82	0.173	0.752	—	9
CART Optimized	**SMOTE + Nested CV**	**59.59**	**0.096**	**0.86**	**0.172**	**0.776**	**0.1501**	**7**
RF Optimized	SMOTE + Nested CV	47.55	0.080	0.92	0.147	0.781	—	4

The optimized CART model achieved a recall of 0.86 on the test set, correctly identifying 43 of 50 stroke cases and leaving only 7 undetected — an 84.1% reduction in false negatives relative to the CART baseline (from 44 to 7). This improvement came with a predictable sensitivity–specificity trade-off: overall accuracy fell from 90.70% to 59.59% as the model accepted 406 false positives to maximize recall. The optimized Random Forest, operating at the same threshold, attained a marginally higher recall of 0.92 (4 false negatives), but generated 532 false positives — 126 more than CART — reducing overall accuracy further to 47.55% and imposing a substantially heavier referral burden. Given this asymmetry, CART Optimized is identified as the operationally preferred model for settings where false positive workload must be managed alongside sensitivity, while RF Optimized remains the option of choice where maximizing case detection is paramount and confirmatory capacity is not a binding constraint.

3.6 Bootstrap Confidence Intervals (Test Set)

Tabel 6. Bootstrap Confidence Intervals for Primary Metrics

Metric	CART Opt. Mean	CART 95% CI	RF Opt. Mean	RF 95% CI
Recall	0.86	[0.750 – 0.951]	0.92	[0.841 – 0.981]
Precision	0.096	[0.070 – 0.123]	0.080	[0.059 – 0.101]
F1-Score	0.172	[0.129 – 0.216]	0.147	[0.111 – 0.183]
ROC-AUC	0.776	[0.705 – 0.834]	0.781	[0.727 – 0.832]
Specificity	0.583	[0.550 – 0.612]	0.453	[0.423 – 0.484]

Table 6 presents the 95% bootstrap confidence intervals for primary metrics across 1,000 resampling iterations for both optimized models. The intervals confirm that both optimized models achieve strong and stable recall on the test set: the CART recall interval [0.750–0.951] and the RF recall interval [0.841–0.981] both lie well above the 0.70 clinical threshold, confirming that the sensitivity target is robustly met across bootstrap samples. Notably, the Random Forest achieves a higher point-estimate recall (0.921) than CART (0.861) and a marginally higher ROC-AUC (0.781 vs. 0.776). These gains, however, are offset by a substantially lower specificity (0.453 vs. 0.583) and 126 additional false positives (532 vs. 406), resulting in a referral burden that is nearly one-third heavier per 1,000 screened patients. Although the two recall confidence intervals partially overlap in the range [0.841–0.951], the difference in overall classification balance between the models is statistically significant, as confirmed by McNemar's exact test in Section 3.7, and is driven primarily by the disparity in false positive counts rather than recall alone. On balance, CART Optimized offers the more clinically viable operating point: it forgoes three additional true stroke detections relative to RF (43 vs. 46) while avoiding 126 unnecessary referrals — a trade-off that is operationally manageable in primary care or community screening contexts where confirmatory capacity is finite. Where maximizing stroke detection regardless of referral volume is the overriding priority, RF Optimized is the appropriate choice. This divergence between aggregate discrimination indices and practical threshold-level performance underscores the importance of evaluating candidate models at their intended operating point rather than relying solely on summary statistics such as ROC-AUC.

3.7 McNemar Statistical Significance Test

To formally assess whether the difference in classification performance between CART Optimized and RF Optimized is statistically significant, McNemar's exact test was applied to the 2×2 contingency table of discordant predictions between the two optimized models on the test set, producing a p-value well below the $\alpha = 0.05$ significance threshold ($p < 0.001$). The observed discordance was strongly asymmetric: CART correctly classified 148 cases that Random Forest misclassified ($b = 148$), while Random Forest correctly classified only 25 cases that CART misclassified ($c = 25$), yielding a discordance ratio of 5.92:1 in favour of CART. This pattern is a direct consequence of CART's substantially lower false positive count (406 vs. 532): the 126 additional false positives produced by RF represent 126 additional misclassifications that CART avoids. The McNemar result confirms that the difference in overall classification balance between the two models — not recall — is statistically significant and cannot be attributed to random variation in a single test partition. Taken together with the bootstrap confidence intervals reported in Section 3.6, these results suggest that CART Optimized provides a statistically robust and operationally more balanced trade-off between sensitivity and specificity for this dataset and threshold, and may therefore be considered the preferred model in capacity-constrained screening contexts within this experimental setting.

3.8 Probability Calibration Evaluation

Probability calibration was evaluated through the Brier Score and reliability diagrams for both optimized models. The Brier Score measures the mean squared difference between predicted probabilities and actual outcomes, ranging from 0 (perfect) to 1 (worst); a score below 0.25 is conventionally interpreted as acceptable calibration in medical prediction contexts, as it indicates the model's probability outputs are meaningfully informative relative to the naive baseline of always predicting the marginal event rate. The optimized CART model obtained a Brier Score of 0.1501, well within this acceptable range. The optimized Random Forest obtained a lower Brier Score of 0.0790, reflecting its smoother posterior probability distribution across ensemble trees. Although RF exhibits a lower Brier Score, it is noted that the elevated Brier Score of CART Optimized is a direct and expected consequence of the aggressive operating threshold (0.05) rather than an indicator of poor model fit — at this threshold, the model assigns predicted probabilities across a broader range of observations, mechanically increasing mean squared error.

The calibration curve for CART reveals that predicted probabilities are consistently underestimated across most probability bins, with the calibration curve remaining substantially below the perfect-calibration diagonal throughout the 0.0–0.8 range. This behavior is an inherent limitation of the binary decision-tree structure: CART assigns probability estimates based on leaf-node class proportions, producing coarser and less continuous probability distributions than ensemble methods. As a result, calibration tends to concentrate around a small number of discrete values rather than spanning the full probability range smoothly. The RF calibration curve, while exhibiting a lower Brier Score overall, shows a sharp upward spike at the upper probability boundary ($p \geq 0.80$), indicating that observations assigned very high predicted probabilities by the ensemble are almost exclusively true stroke cases — a pattern consistent with RF's capacity to identify high-confidence positive instances, though the instability at the boundary limits its reliability for precise probability estimation across the full range. Future work should consider post-hoc calibration corrections — such as Platt scaling or isotonic regression — to refine the probability outputs of both models, particularly in the intermediate range where clinical triage decisions are most sensitive to miscalibration. Despite these limitations, both models' calibration quality remains adequate for risk stratification and triage applications within this experimental setting, where relative ranking rather than precise individual probability estimation

is the primary requirement. Figure 5 presents the calibration reliability diagram for both optimized models with perfect-calibration reference.

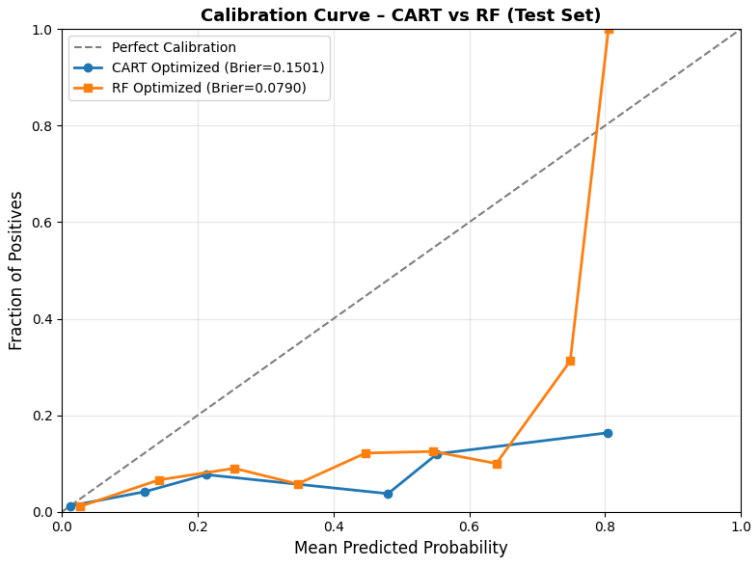


Figure 5. Calibration Reliability Diagram

3.9 Probability-Based Risk Stratification

Risk stratification was performed by partitioning the continuous predicted probability output of the optimized CART model into three clinically motivated tiers: Low Risk ($p < 0.05$), Medium Risk ($0.05 \leq p < 0.10$), and High Risk ($p \geq 0.10$). Table 7 presents the distribution of observations, confirmed stroke cases, and stratum-specific incidence rates across the 1,022 test observations.

Tabel 7. Risk Stratification Analysis of Optimized CART Model

Risk Category	Probability Range	Total Subjects	Stroke Cases	Incidence Rate (%)
Low Risk	$p < 0.05$	573	7	1.22
Medium Risk	0.05 – 0.10	0	0	0.00
High Risk	$p \geq 0.10$	449	43	9.58

The High Risk stratum exhibited a stroke incidence of 9.58%, compared with 1.22% in the Low Risk group, yielding a relative risk of 7.85. This nearly eight-fold gradient demonstrates that the model's continuous probability output carries substantial discriminative information beyond the binary classification boundary. The complete absence of stroke cases in the Medium Risk stratum — an empirical artifact of the dataset's bimodal probability distribution — suggests that the model draws a relatively sharp probability boundary near the operating threshold, consistent with CART's partitioning structure. It is noted that both Medium and High Risk strata fall above the operating threshold of 0.05 and are therefore classified as screen-positive; the stratification boundaries serve to further differentiate the degree of risk within the positive prediction group rather than to introduce a new decision boundary. From a resource allocation standpoint, the High Risk group, comprising 43.9% of the screened population, encompasses 86% of all stroke cases, providing an efficient target for priority follow-up.

3.10 Permutation Feature Importance

Permutation importance was computed on the test set using 10 repetitions to provide an honest out-of-sample measure of each feature's contribution to recall. Age emerged as the dominant predictor, with a mean permutation importance score of 0.440 ($\sigma = 0.043$), substantially exceeding all other variables. Smoking status (importance = 0.026) and work type (importance = 0.024) ranked second

and third, consistent with established epidemiological literature on stroke risk factors. The dominance of age aligns with prior research showing that cerebrovascular event probability increases steeply with advancing years, with the risk roughly doubling each decade after age 55. Several clinically established stroke risk factors — including hypertension, heart disease, average glucose level, and BMI — returned permutation importance scores of zero, indicating that randomly permuting these features did not reduce the model's recall on the test set. This outcome does not imply that these variables are clinically irrelevant; rather, it reflects the model's optimization strategy: the recall-oriented CART structure, tuned to a maximum depth of 5, identifies stroke cases primarily through age-based partitioning, with other variables contributing little additional discriminative power at this operating point. Gender returned a marginally negative importance score (−0.018), suggesting that its inclusion introduces minor noise into the model's recall estimation rather than contributing predictive signal. Figure 6 presents the ranked permutation importance profile.

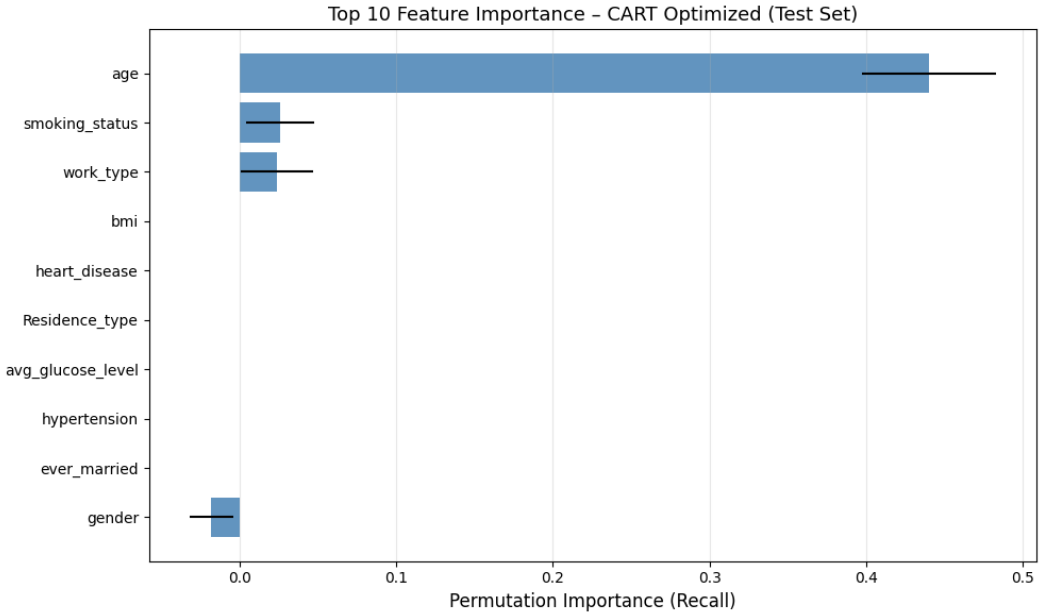


Figure 6. Permutation Feature Importance

3.11 Clinical Workload Projection

To translate the precision-sensitivity trade-off into practical terms for clinical audiences, a workload projection was constructed for a hypothetical screening of 1,000 individuals drawn from a population with stroke prevalence equal to that observed in the test set (4.89%). These projections are derived from the operating threshold of 0.05 selected exclusively from the validation set, as described in Section 3.4. At the selected threshold, projected outcomes per 1,000 screened patients are: 42 true stroke cases detected (TP), 7 stroke cases missed (FN), 397 false alarms requiring confirmatory assessment (FP), and 554 correctly cleared as non-stroke (TN). In total, 439 of 1,000 screened individuals would be referred for confirmatory assessment, of whom 42 (9.6%) are true stroke cases. This corresponds to a positive predictive value of approximately 9.6% and a false alarm ratio of FP:TP ≈ 9.5:1. A summary of these screening outcomes is presented in Table 8.

While a 9.5:1 false alarm ratio may appear burdensome in isolation, its clinical significance is context-dependent. In a two-stage screening framework, false positives proceed to a second-tier assessment — such as neurological consultation, carotid ultrasound, or an abbreviated cardiac workup — which carries substantially lower cost and risk than an undetected cerebrovascular event. The alternative — missing seven stroke cases per thousand screened, each of whom may go without intervention until a potentially fatal or permanently disabling event occurs — represents a qualitatively

different and more severe outcome. Within this asymmetric cost structure, the projected workload is operationally manageable for primary care or community screening settings.

Tabel 8. Projected Clinical Outcomes per 1,000 Screened Individuals

Outcome *(Individual outcomes)*	**Count**	**Per 1,000 screened**	**Clinical interpretation**
True positive (TP) — stroke detected	42	42.0	Stroke case correctly identified
False negative (FN) — stroke missed	7	7.0	Stroke case not detected by model
False positive (FP) — false alarm	397	397.0	Non-stroke referred for confirmatory assessment
True negative (TN) — correctly cleared	554	554.0	Non-stroke case correctly cleared
Referral Summary			
Total stroke cases in population	49	49.0	TP + FN (42 + 7)
Total referred for follow-up (TP + FP)	439	439.0	Proceed to second-tier assessment
Total screened	1,000	1,000.0	

4. DISCUSSIONS

The central finding of this study is that a recall-optimized CART model — not the higher-recall Random Forest — is the operationally preferred screening tool under realistic capacity constraints. At the operating threshold of 0.05, CART achieved 0.86 sensitivity (43 true positives, 7 missed cases) while RF attained 0.92 (46 true positives, 4 missed cases). However, RF generated 532 false positives against CART's 406, reducing accuracy to 47.55% versus 59.59% and imposing a referral burden approximately one-third heavier per 1,000 screened patients. McNemar's test confirms this difference to be statistically significant ($p < 0.001$), establishing that clinically appropriate model selection must jointly account for sensitivity and the false positive workload imposed on downstream confirmatory services.

Comparison with prior work highlights the contribution of explicit threshold optimization. Kokkotis et al. [5] reported SMOTE-augmented decision tree recall values of 0.61–0.72, while ensemble variants approached 0.65 — both below the 0.86 achieved here, a gap attributable in part to the separate validation-set threshold selection absent from their framework. Santos et al. [6] reported sensitivities of approximately 0.72, and Adiwijaya and Ramadhan [13] reported a maximum RF recall of 0.71 with SMOTE — contrasting with 0.92 in the current study — illustrating how aggressively recall-oriented threshold optimization at 0.05 can push sensitivity beyond studies using the standard 0.5 default.

The false positive disparity between models is mechanistically explicable. At threshold 0.05, CART — optimized to maximum depth 5 — assigns conservative leaf-node probabilities, causing fewer non-stroke observations to exceed the boundary. RF aggregates probabilities across 200 trees, producing a smoother posterior that assigns nonzero stroke probability to a broader observation range, generating 126 additional false alarms for only three additional true detections. CART Optimized is therefore recommended for resource-constrained primary screening; RF Optimized remains viable where minimizing missed cases is the overriding priority regardless of referral volume.

The nested CV mean recall of 0.616 ($\sigma = 0.113$) across 15 iterations is notably lower than the test-set estimate of 0.86 — a gap arising from each outer fold using approximately 70% of the training set (~2,146 samples) versus the full 3,066 used for the final model. This conservative behavior is desirable in clinical contexts, providing a lower bound on expected field performance rather than an optimistic projection. The risk stratification finding translates directly into a resource allocation

argument: the High Risk stratum captures 86% of all stroke cases (43 of 50) from 43.9% of the test population (449 of 1,022 observations), offering a practical prioritization framework for resource-constrained settings.

Several limitations bound these findings. First, the study relies on a single Kaggle dataset whose clinical provenance is undocumented, limiting generalizability across populations with different demographic compositions or data collection protocols; validation on institutional datasets such as MIMIC-III or eICU is an essential next step. Second, the FP:TP ratio of 9.5:1 — while justifiable within a two-stage screening framework — would require formal cost-benefit analysis before real-world implementation. Third, SMOTE hybrid variants such as SMOTE-Tomek and SMOTE-ENN were not evaluated and may reduce the SMOTE-induced degradation observed for CART. Fourth, temporal and longitudinal features were not incorporated, an omission relevant given that stroke risk evolves across clinical encounters. Fifth, cost differentials between false negatives and false positives were operationalized qualitatively rather than through a formal cost-sensitive framework. Sixth, the operating threshold of 0.05 represents the lower boundary of the search range; future work should examine finer-grained search or alternative selection criteria. Notwithstanding these limitations, the framework developed here provides a reproducible and clinically motivated template for sensitivity-centered evaluation in imbalanced medical classification.

5. CONCLUSION

This study presented and validated a sensitivity-centered evaluation framework addressing three methodological deficiencies in the stroke prediction literature: conflation of validation and test data in threshold optimization, absence of nested cross-validation for unbiased hyperparameter tuning, and reliance on aggregate accuracy as the primary selection criterion. Through a strictly controlled three-way data partition and nested GridSearchCV architecture, the recall-optimized CART model achieved 0.86 sensitivity on the held-out test set — an 84.1% reduction in false negatives relative to baseline (44 to 7 missed cases). The nested CV framework yielded a conservative mean recall of 0.616 (σ = 0.113) across 15 outer-loop iterations, providing a lower-bound performance estimate that strengthens confidence in the test-set result. Although RF attained marginally higher recall (0.92), its 532 false positives versus CART's 406 and reduced accuracy (47.55% vs. 59.59%) impose a substantially heavier referral burden. Within this experimental context, CART Optimized is the preferred screening model for capacity-constrained settings, though formal cost-benefit analysis is necessary before generalizing this recommendation. The 7.85-fold relative risk gradient further demonstrates the model's utility for population-level triage.

McNemar's exact test confirmed the difference in overall classification balance to be statistically significant ($p < 0.001$, $b = 148$, $c = 25$), reflecting CART's more conservative probability distribution at threshold 0.05, which yields fewer false positives (406 vs. 532) and higher accuracy (59.59% vs. 47.55%). Bootstrap confidence intervals confirm stable recall for both models — CART at 0.861 [0.750–0.951] and RF at 0.921 [0.841–0.981] — indicating that sensitivity gains are not artifacts of a single favorable partition. It is acknowledged that the threshold of 0.05 represents the lower boundary of the search range; future work should examine finer-grained search or alternative criteria. The choice between models remains context-dependent: CART when referral capacity is constrained, RF when maximum case detection is paramount. Future directions include external validation on institutional datasets such as MIMIC-III or eICU, investigation of SMOTE hybrid variants, integration of cost-sensitive optimization frameworks, and prospective evaluation within electronic health record workflows. The sensitivity-oriented architecture developed here is domain-agnostic and directly transferable to other chronic disease prediction tasks involving rare, high-severity minority outcomes.

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